The Least Squares Method - Part 2

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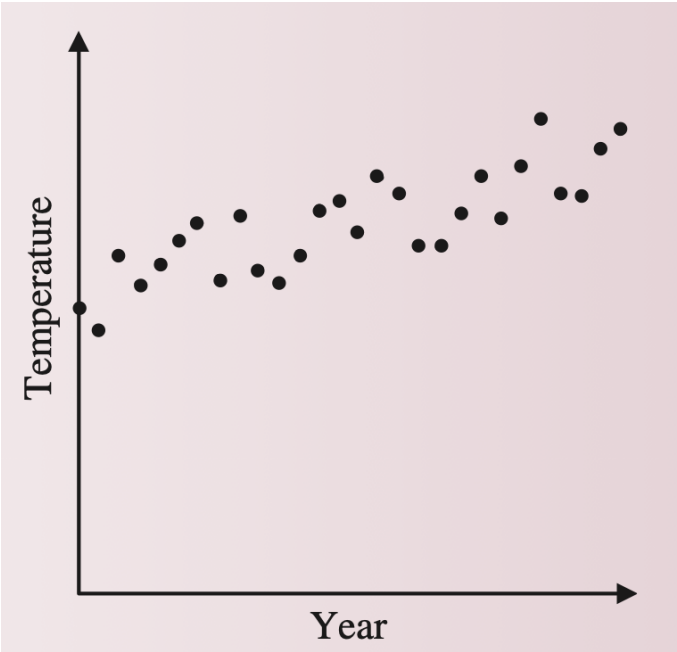
Real Life Example

- suppose we are given the average temperature of the Earth's surface from 1975 through 2002;
- can we infer about the trend of the Earth's temperature? can we estimate the Earth's temperature for 2023?

Average Global Temperatures 1975–2002					
1975	13.94	1985	14.03	1994	14.25
1976	13.86	1986	14.12	1995	14.37
1977	14.11	1987	14.27	1996	14.23
1978	14.02	1988	14.29	1997	14.40
1979	14.09	1989	14.19	1998	14.56
1980	14.16	1990	14.37	1999	14.32
1981	14.22	1991	14.32	2000	14.31
1982	14.04	1992	14.14	2001	14.46
1983	14.25	1993	14.14	2002	14.52
1984	14.07				

Introduction to Linear Algebra with Applications, DeFranza & Gagliardi



Last lecture...

Normal Equations: Numerical Instability

$$\tilde{x} = \left(A^T A\right)^{-1} A^T b$$

the problem: increased numerical instability in $A^T A$

example: with $\eta < \varepsilon \leq \sqrt{\eta}$,

$$A = \begin{bmatrix} 1 & 1 & 1 \\ \varepsilon & 0 & 0 \\ 0 & \varepsilon & 0 \\ 0 & 0 & \varepsilon \end{bmatrix}, \qquad A^T A = \begin{bmatrix} 1+\varepsilon^2 & 1 & 1 \\ 1 & 1+\varepsilon^2 & 1 \\ 1 & 1 & 1+\varepsilon^2 \end{bmatrix}.$$

Normal Equations: Numerical Instability

If A is square:

$$\frac{\|\mathbf{x}-\tilde{\mathbf{x}}\|}{\|\mathbf{x}\|} \leq \|\mathbf{A}\| \left\|\mathbf{A}^{-1}\right\| \frac{\|\tilde{\mathbf{r}}\|}{\|\mathbf{b}\|}, \qquad \kappa(A) = \|\mathbf{A}\| \left\|\mathbf{A}^{-1}\right\| = \frac{\lambda_{\max}}{\lambda_{\min}};$$

If A is rectangular:

$$\frac{\|\mathbf{x}-\tilde{\mathbf{x}}\|}{\|\mathbf{x}\|} \leq \|\mathbf{A}\| \left\|\mathbf{A}^{+}\right\| \frac{\|\tilde{\mathbf{r}}\|}{\|\mathbf{b}\|}, \qquad \kappa(A) = \|\mathbf{A}\| \left\|\mathbf{A}^{+}\right\| = \frac{\sigma_{\max}}{\sigma_{\min}}.$$

In least squares, the relevant matrix is $B = A^T A$:

$$\kappa_2(B) = \frac{\lambda^B_{\max}}{\lambda^B_{\min}} = \frac{(\sigma^A_{\max})^2}{(\sigma^A_{\min})^2} = \kappa_2^2(A)$$

We want an algorithm whose error bound depends on $\kappa_2(A)$ and not $\kappa_2^2(A)$.

Least Squares Method via QR Decomposition

- if $A_{m\times n}$ if full column rank, then $A = QR$, where $Q_{m\times n}$ is $Q^T Q = I$ and $R_{n\times n}$ is a non-singular upper triangular;
- then,

$$\begin{aligned}
\tilde{x} &= \left(A^T A\right)^{-1} A^T b \\
&= \left(R^T Q^T Q R\right)^{-1} R^T Q^T b \\
&= \left(R^T R\right)^{-1} R^T Q^T b \\
&= R^{-1} \left(R^T\right)^{-1} R^T Q^T b \\
&= R^{-1} Q^T b
\end{aligned}$$

- solve for $\tilde{x}$ from $R\tilde{x} = Q^T b$, with backward substitution.

Ill-conditioning: Least Squares via SVD

- if the columns of A are not "sufficiently" linearly independent, the backward pass on R will fail due to very small magnitude (or zero) diagonal entries;
- instead of $A = QR$, the useful factorization of A is SVD: $A = U\Sigma V^T$.

Today's lecture...

Data Fitting Problem

- we are given some data

$$x^{(1)}, \ldots, x^{(N)}, \quad y^{(1)}, \ldots, y^{(N)}$$

  also called observations, samples, or measurements;
- $(x^{(i)}, y^{(i)})$ is the i-th data pair;
- we believe x and y are related by an (unknown) model

$$y \approx f(x);$$

- $x \in \mathbb{R}^n$ is the independent variable or feature vector;
- $y \in \mathbb{R}$ is the response variable, corresponding to something we want to predict;
- goal: build a model $\hat{f}$ that approximates f.

Model

- choose model $\hat{f} : \mathbb{R}^n \to \mathbb{R}$ linear in the parameters θ_i $(i = 1, \ldots, p)$:

$$\hat{f} = \theta_1 f_1(x) + \cdots + \theta_p f_p(x);$$

- $f_i(x) : \mathbb{R}^n \to \mathbb{R}$ are *basis functions* we choose;
- θ_i are p *model parameters* that we need to compute;
- $\hat{y}^{(i)} = \hat{f}(x^{(i)})$ is the model's prediction of $y^{(i)}$;
- we would like $\hat{y}^{(i)} \approx y^{(i)}$, i.e., the model is consistent with the observed data.

Least Squares Data Fitting

- prediction error or residual is $r^{(i)} = y^{(i)} - \hat{y}^{(i)}$;
- least squares data fitting: find model parameters θ_i to minimize the root mean square (RMS) prediction error on data set, that is,

$$\min_{\theta_1,\theta_2,\ldots,\theta_p} \left(\sum_{i=1}^{N} (r^{(i)})^2 / N \right)^{1/2};$$

- this can be formulated and solved as a least squares problem:

$$\min_{\theta_1,\theta_2,\ldots,\theta_p} \sum_{i=1}^{N} (r^{(i)})^2;$$

Least Squares Data Fitting

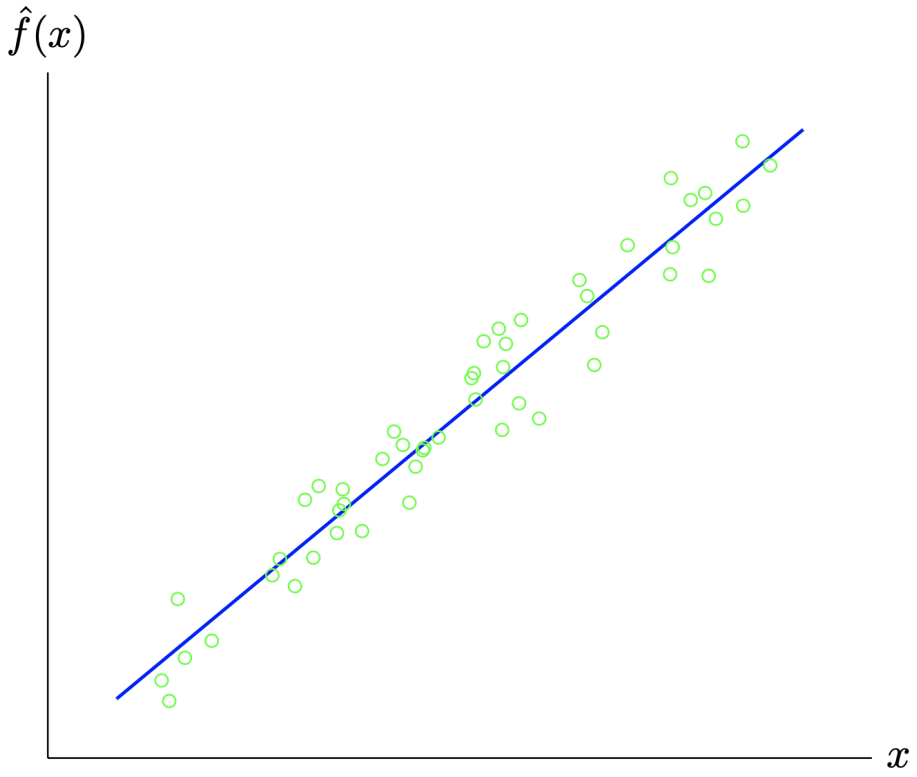
- express $y^{(i)}$, $\hat{y}^{(i)}$, and $r^{(i)}$ as $N \times 1$ vectors:
  - $y^{\mathrm{d}} = (y^{(1)}, \ldots, y^{(N)})$ is vector of outcomes ;
  - $\hat{y}^{\mathrm{d}} = (\hat{y}^{(1)}, \ldots, \hat{y}^{(N)})$ is vector of predictions;
  - $r^{\mathrm{d}} = (r^{(1)}, \ldots, r^{(N)})$ is vector of residuals;
- define $N \times p$ matrix A with elements $A_{ij} = f_j(x^{(i)})$, so $\hat{y}^{\mathrm{d}} = A\theta$;
- least squares data fitting: find $\hat{\theta}$ to minimize

$$||r^{\mathrm{d}}||^2 = ||y^{\mathrm{d}} - \hat{y}^{\mathrm{d}}||^2 = ||y^{\mathrm{d}} - A\theta||^2$$

  with respect to θ;
- $\hat{\theta} = (A^T A)^{-1} A^T y^{\mathrm{d}}$ (if columns of A are linearly independent);
- $||A\hat{\theta} - y^{\mathrm{d}}||^2/N$ is the mean-square fitting error;

Fitting Univariate Functions

- when $n = 1$, we seek to approximate a function $f : \mathbb{R} \to \mathbb{R}$;
- we can plot the data $(x^{(i)}, y^{(i)})$ and guess the model function $\hat{y} = \hat{f}(x)$:



Line Fit

- $p = 2$, with $f_1(x) = 1$, $f_2(x) = x$;
- model has the form $\hat{f}(x) = \theta_1 + \theta_2 x$;
- matrix A has the form

$$\begin{bmatrix} 1 & x^{(1)} \\ 1 & x^{(2)} \\ \vdots & \vdots \\ 1 & x^{(N)} \end{bmatrix}$$

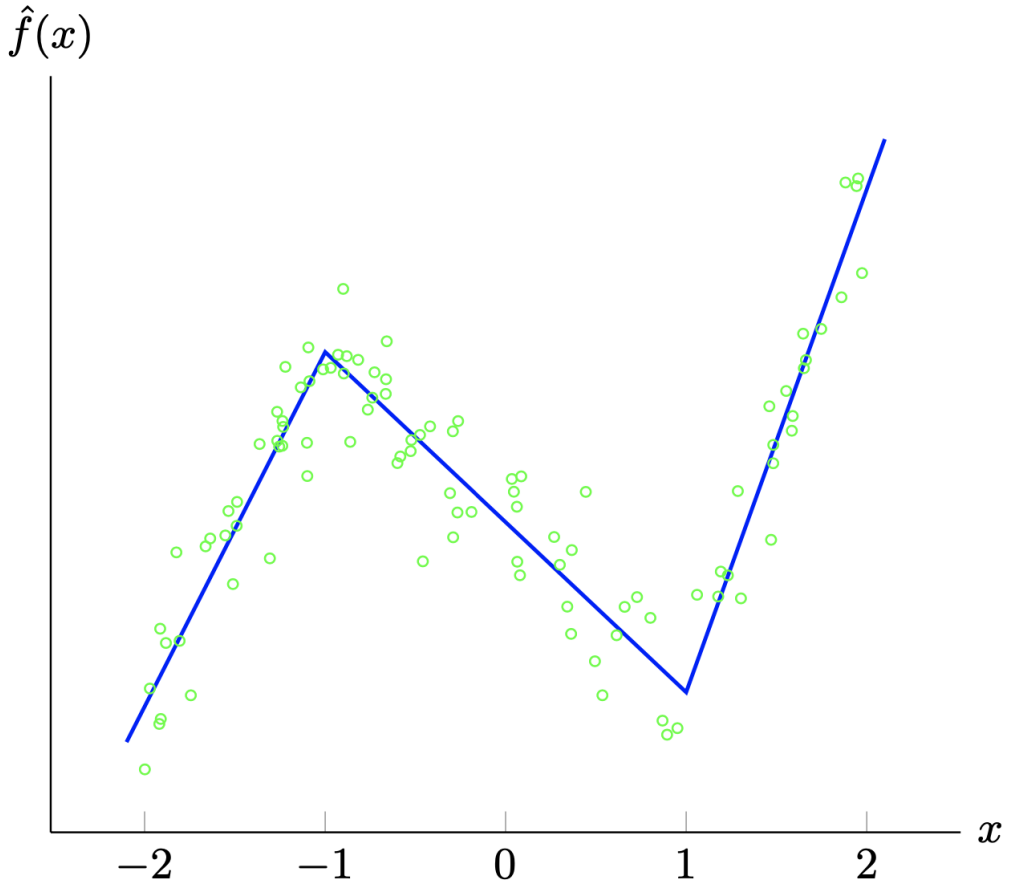
- $A^T A\theta = A^T y^{\mathrm{d}}$ can be written explicitly as a 2×2 system:

$$\begin{bmatrix} \mathbf{1}^T \mathbf{1} & \mathbf{1}^T x^{\mathrm{d}} \\ \mathbf{1}^T x^{\mathrm{d}} & (x^{\mathrm{d}})^T x^{\mathrm{d}} \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} \mathbf{1}^T y^{\mathrm{d}} \\ (x^{\mathrm{d}})^T y^{\mathrm{d}} \end{bmatrix},$$

in which $\mathbf{1}$ is the ones vector in $\mathbb{R}^N$.

Piecewise Linear Fit

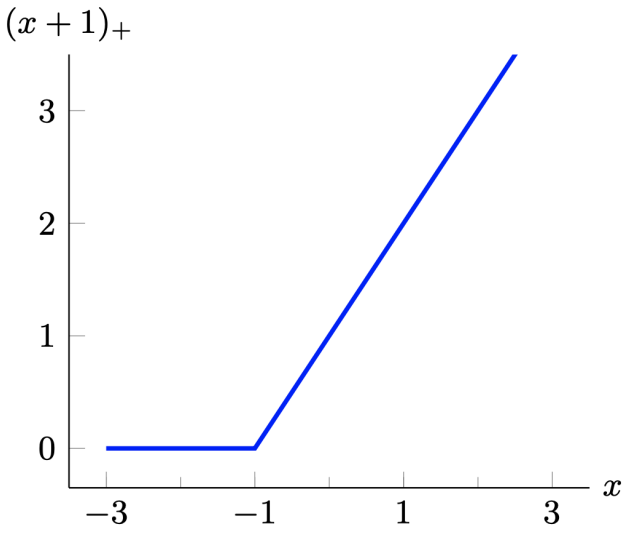
- plot data and check for knot points;
- for the data below, knot points at $x = -1$ and $x = +1$:

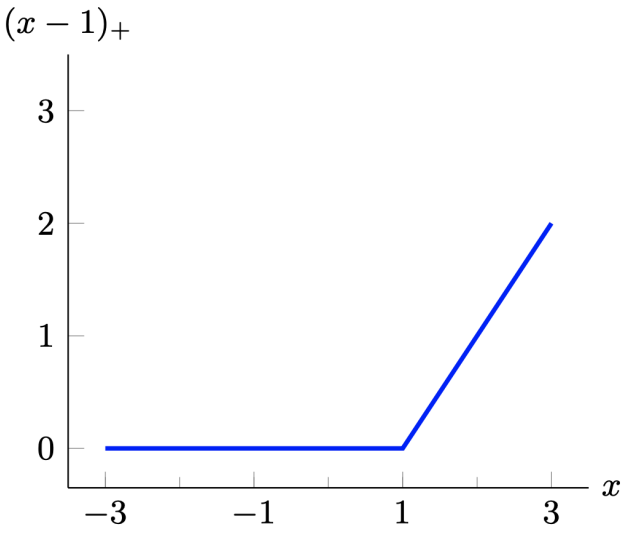


Piecewise-linear Basis Functions

- we can describe any piecewise-linear function with a finite number of knots points;
- augment the previous set of basis functions, $\{1, x\}$, with additional functions of the form

$$(x-a)_+ \text{ where } (u)_+ = \max(u, 0) :$$





Polynomial Fit

- $f_i(x) = x^{i-1}$, $i = 1, \ldots, p$;
- the model is a polynomial of degree than p:

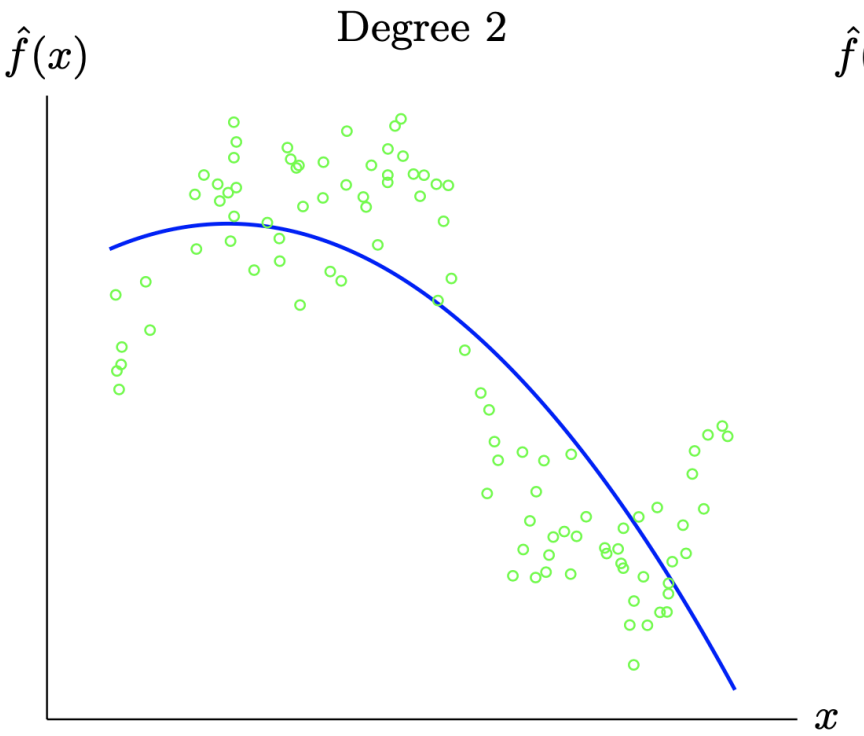
$$\hat{f}(x) = \theta_1 + \theta_2 x + \cdots + \theta_p x^{p-1};$$

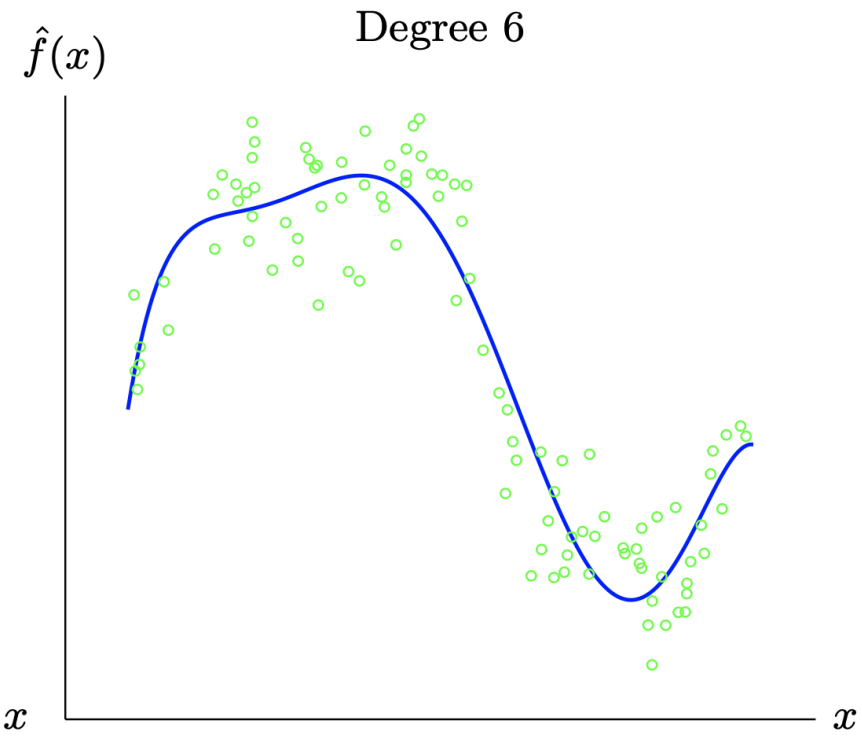
  (x^i means x raised to the power i, $x^{(i)}$ is ith data point)
- A is the matrix

$$A = \begin{bmatrix} 1 & x^{(1)} & \ldots & (x^{(1)})^{p-1} \\ 1 & x^{(2)} & \ldots & (x^{(2)})^{p-1} \\ \vdots & \vdots & & \vdots \\ 1 & x^{(N)} & \ldots & (x^{(N)})^{p-1} \end{bmatrix};$$

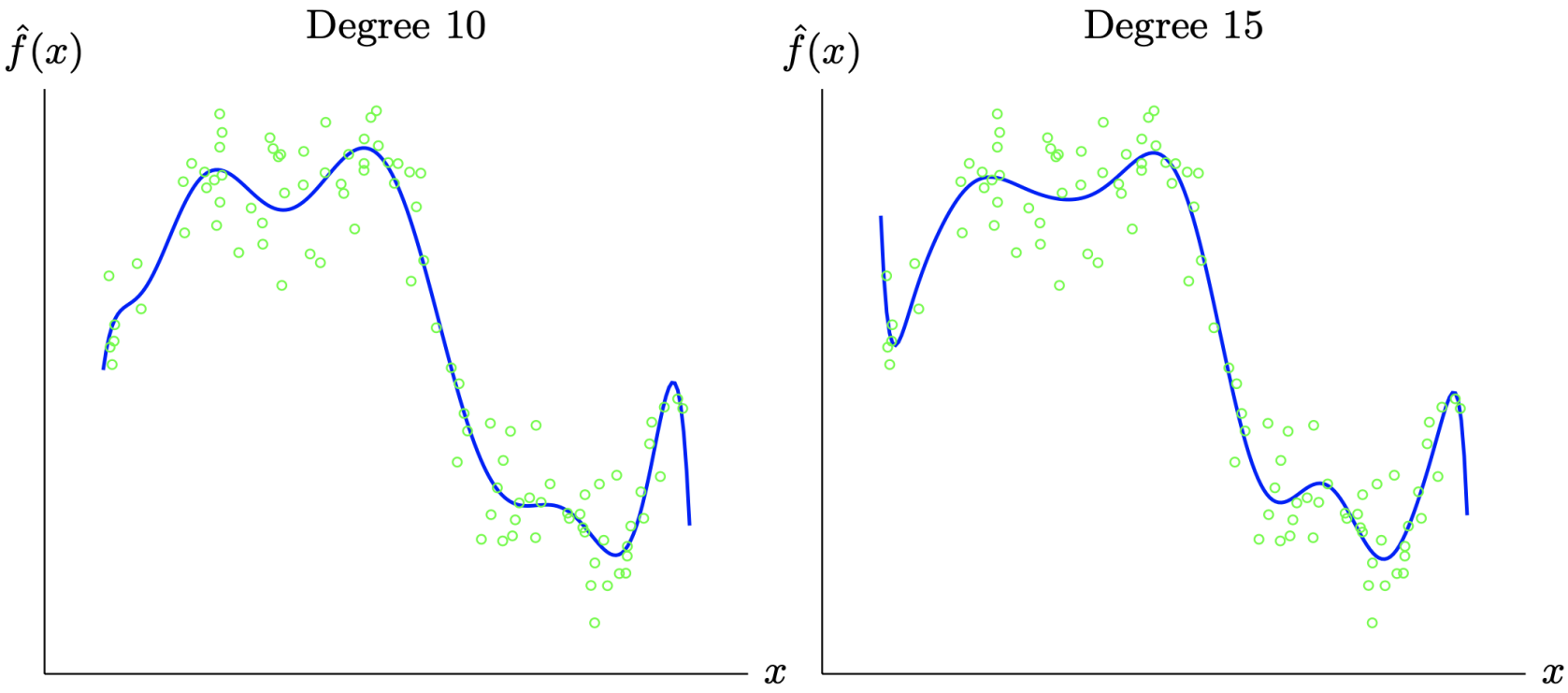
Polynomial Fit

$N = 100$ data points





Overfitting



- do these solutions have lower RMS prediction error?
- should they be used? careful not to overfit;

Overfitting: fitting the noise instead of the trend

- as the poly degree increases, the model becomes more flexible;
- this usually reduces the training RMS error, because the curve can pass closer to the data points;
- but a very flexible model may start fitting not only the underlying pattern, but also the random noise in the sample;
- this is overfitting.

Key point:

A model can have a very small error on the training data and still perform poorly on new, unseen data:

$$\text{training error} \downarrow \quad \not\Rightarrow \quad \text{prediction error} \downarrow$$

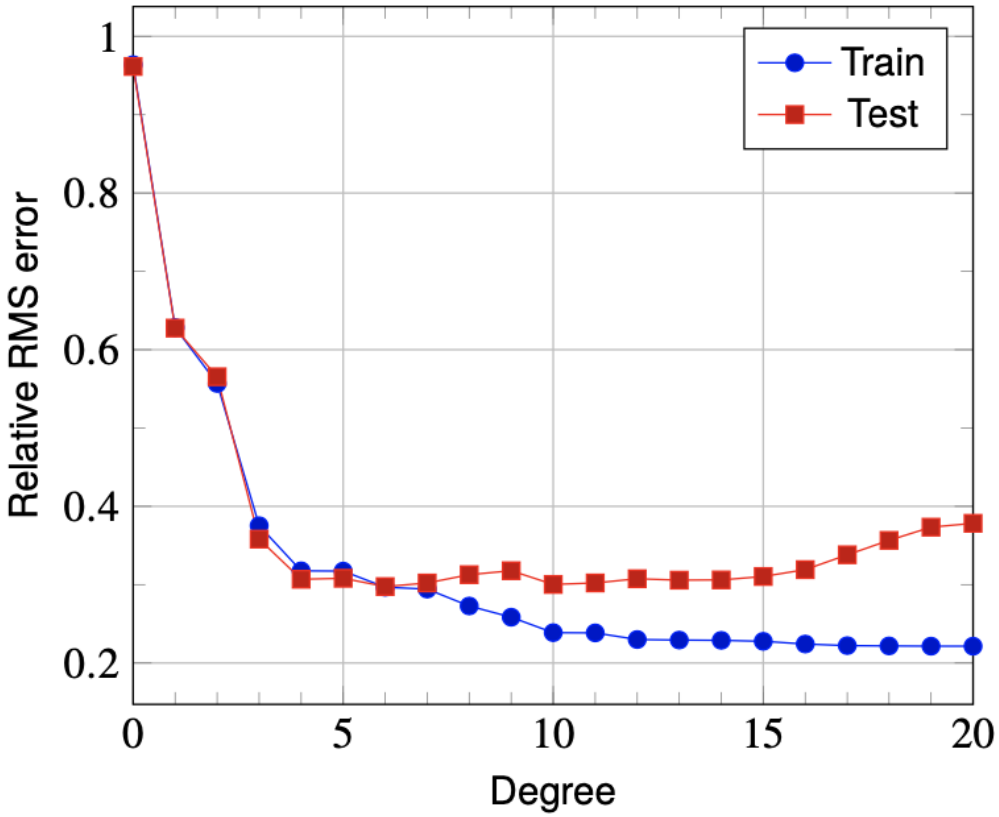
- low degree: may underfit (too rigid);
- very high degree: may overfit (too wiggly);
- good modeling seeks the best balance between fit and generalization.

Validation

- how to choose among different candidate models? that is, how to choose among polynomials of different degrees?
- divide the data set into a *training* set and a *test* set;
- the training set is used to find the best θ;
- if the test RMS prediction error is large, the model is not predicting well;
- a complex model with large test RMS prediction error signals overfitting of the data set.

RMS Error on Test Data

- degree 4, 5 or 6 are reasonable choices:



Example

- suppose we are given the number of bike rentals at a Seoul location for almost 9000 days, and the corresponding daily temperature, humidity, rainfall and windspeed;
- if we have today's weather variables, can we estimate the number of bike rentals at that same location?

Linear Polynomial Model with Two Features: $n = 2$

- until now $x^{(i)} \in \mathbb{R}$;
- suppose we are give a set of N points in the plane:

$$x^{(1)}, \ldots, x^{(N)} \in \mathbb{R}^2 \qquad y^{(1)}, \ldots, y^{(N)} \in \mathbb{R};$$

- we want to fit a *plane* through these points:

$$\hat{f} = \theta_1 + \theta_2 x_1 + \theta_3 x_2;$$

- the basis functions are now

$$f_1(x) = 1, \quad f_2(x) = x_1, \quad f_3(x) = x_2;$$

- A is of size $N \times 3$, $A_{ij} = f_j(x^{(i)})$.

Polynomial Model with Multiple Features: $n > 1$

- fit a degree-2 polynomial with n features:

$$\hat{f}(x) = a + b^T x + x^T C x, \quad x \in \mathbb{R}^n;$$

- assume $n = 2$ and symmetric C, for simplicity;
- given data set $x^{(i)} \in \mathbb{R}^2$ and corresponding $y^{(i)}$;
- we can expand the fit as:

$$\hat{f}(x) = a + b_1 x_1 + b_2 x_2 + c_{11} x_1^2 + c_{22} x_2^2 + 2 c_{12} x_1 x_2;$$

- $p = 6$: 6 basis functions and 6 θ_i parameters:

$$f_1(x) = 1, \; f_2(x) = x_1, \; f_3(x) = x_2,$$

$$f_4(x) = x_1^2, \; f_5(x) = x_2^2, \; f_6(x) = 2 x_1 x_2;$$

- A is of size $N \times 6$, $A_{ij} = f_j(x^{(i)})$.

Summary

- least squares problems appear frequently in applications;
- modern data science and analytics relies and builds on these ideas.